

Stephen Sun

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EXPERIENCE

Microsoft AI

Mar 2025 - Present

Software Engineer II - Bing Maps Data Infra team

Redmond, WA

- Primary developer in migration of Bing Maps team's data pipelines from legacy proprietary tools to open-source frameworks such as Apache Spark, Iceberg, and Airflow. Improved pipeline performance by 50% through migration.
- Built validation suite in Scala and wrote tests to verify map data integrity before export to downstream teams.
- Implemented redesign of team's primary geospatial data store to store rich data attributes such as business reviews.

ServiceNow

Jul 2022 - Mar 2025

Software Engineer - Big Data team

Santa Clara, CA

- Built and managed data pipelines in Apache Spark to enable data query and visualization of ServiceNow customer data.
- Developed framework to parse raw log files, then correlate them with alerts and anomalies. Support and Site Reliability teams' Mean Time to Recover was improved from >1h to 1min per engineer, and >1000 company hours are saved yearly.
- Deployed key ServiceNow global instance availability data pipeline with minute-level data and daily and monthly aggregations. Data is used by 200+ users through real-time visualizations and aggregated reports.
- Onboarded critical pipeline to ingest user experience data from >20,000 customers for usage and performance analytics.

Brown University

Sep 2021 - Dec 2021

Teaching Assistant - CSCI 1600: Real-Time and Embedded Software

Providence, RI

- Developed course material for TA-led labs with Prof Milda Zizyte for the inaugural run of the class, with 35 students.
- Ran labs using Arduino MKR1000, mentored student groups for a semester long design project and graded homework.

ServiceNow

May 2021 - Aug 2021

Software Engineer Intern - Big Data team

Santa Clara, CA

- Primary developer for a logging framework for BigData team's MariaDB database to track all CRUD operations. Framework is still regularly used today for tracking changes and debugging.
- Created Jenkins job to regularly check 6000+ metrics for discrepancies in metadata, then alert engineers for resolution.
- Participated in ServiceNow's internal hackathon and was part of the top 8 finalists out of over 50 teams.

EDUCATION

Brown University

Sep 2018 - May 2022

Bachelor of Science: ScB - Computer Science and AB - Engineering; GPA: 3.86/4

Providence, RI

- Coursework:** CS - Operating Systems, Machine Learning, Data Science, Programming Languages, Distributed Systems Engineering - Design of Robotic Systems, Digital Electronics System Design, Design of Computing Systems
- Leadership:** President - Brown University Merlions (Singapore Student Club), Co-Founder/VP - Brown Flying Club

PROJECTS

Undergraduate Teaching and Research Award - CubeSat Robot Arm

- Built proof-of-concept robotic arm for use on nanosatellites under Brown School of Engineering's Prof Rick Fleeter. Arm was designed in SolidWorks, control software was coded in MATLAB and Arduino using Inverse Kinematics concepts.
- Prototype arm and software serves as the foundation for the Brown Space Engineering student club's PVDX CubeSat, and project was featured on Brown University's news portal.

Independent Study Course Project - Indoor Robot Navigation with 3D LiDAR

- Programmed TurtleBot 2 in C++ to autonomously navigate through an indoor environment using Simultaneous Location and Mapping (SLAM) and people detection algorithms, while equipped with a Velodyne VLP-16 LiDAR.

Helios - Smart Blinds and Alarm Clock

- Built and programmed wireless voice-controlled smart blinds using Arduino Uno and BitVoicer for speech recognition.

Weenix OS - Barebones Unix Operating System

- Constructed Unix-like operating system in C that is able to make system calls, run system commands and user code.

SKILLS

Languages: Scala, Java, Python, SQL, Bash, C, C++, Arduino, MATLAB

Technologies: Apache Spark, Iceberg, Airflow, Kafka, Impala, Parquet, Kudu, HDFS, HBase, Git, Cloudera Data Platform, Google Cloud Platform, Microsoft Azure, Azure Data Factory, Azure Synapse Analytics, InfluxDB, Grafana